

# Bayesian Graphical Modeling

## Theory

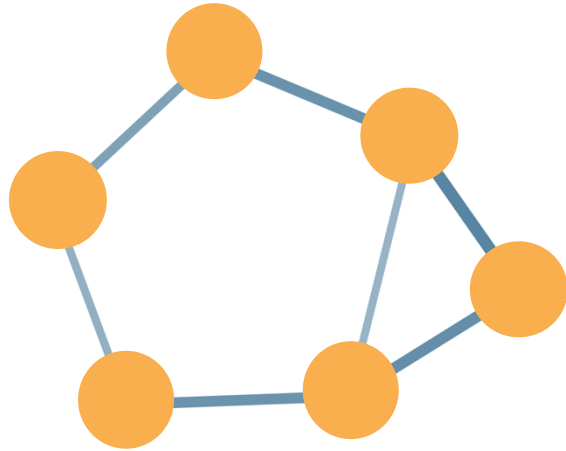
*PsyNets Winterschool 2026*

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[bayesiangraphicalmodeling.com](http://bayesiangraphicalmodeling.com)

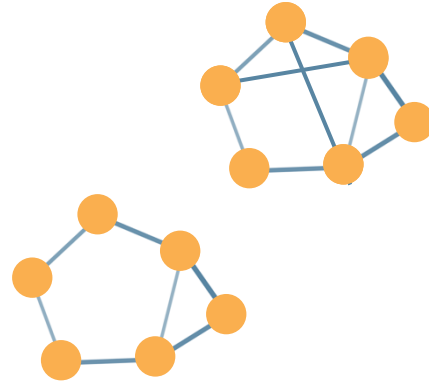


# Setup Today



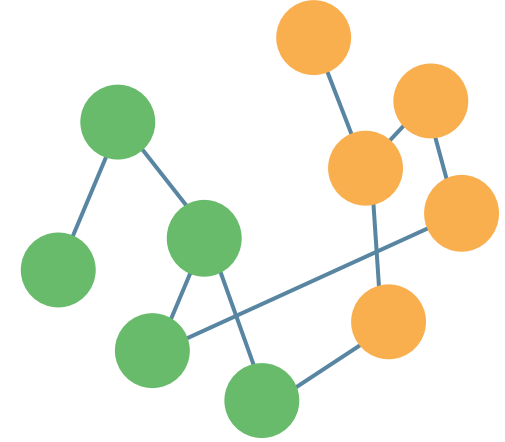
Estimating & Testing

Morning Session



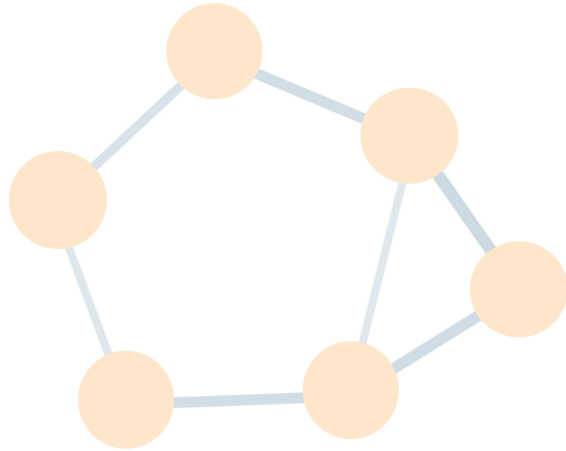
Network Comparison

Afternoon Session



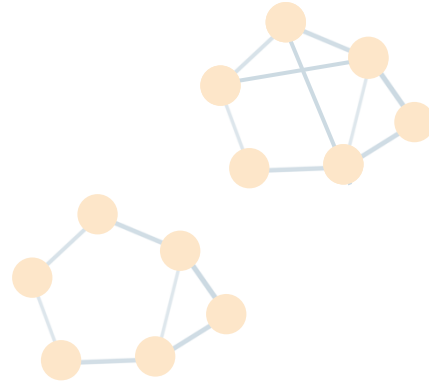
Clustering

# Setup Today



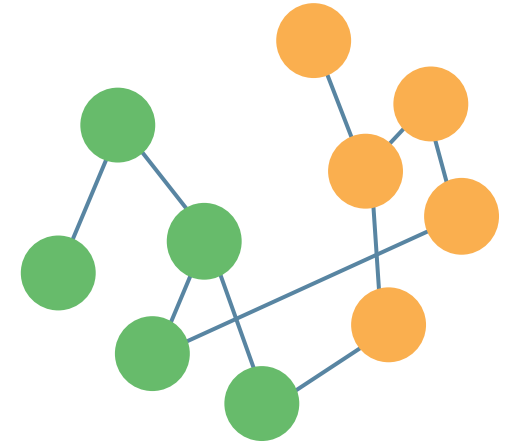
Estimating & Testing

Morning Session



Network Comparison

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Clustering

# Session setup



Part I: Theory



Part II: Practical



Part III: Project

# Priors on the network structure

The prior on the structure,  $p(\mathbf{G})$ , says which network structures we consider plausible before seeing the data.

Specifying it is one of the defining features of Bayesian graphical modeling.

# Two structure priors so far

**Bernoulli** — each edge is included with a fixed and independent probability.

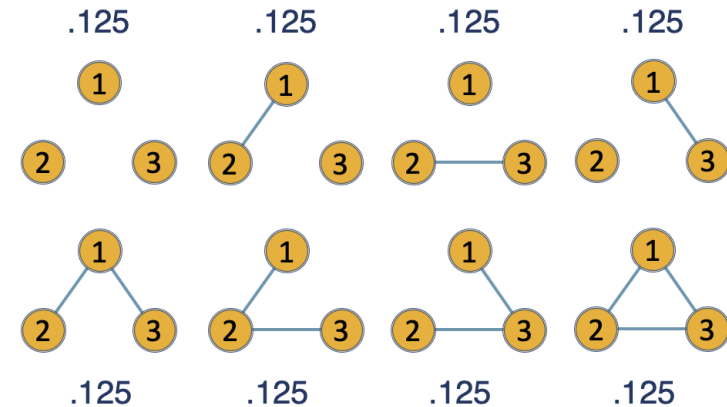
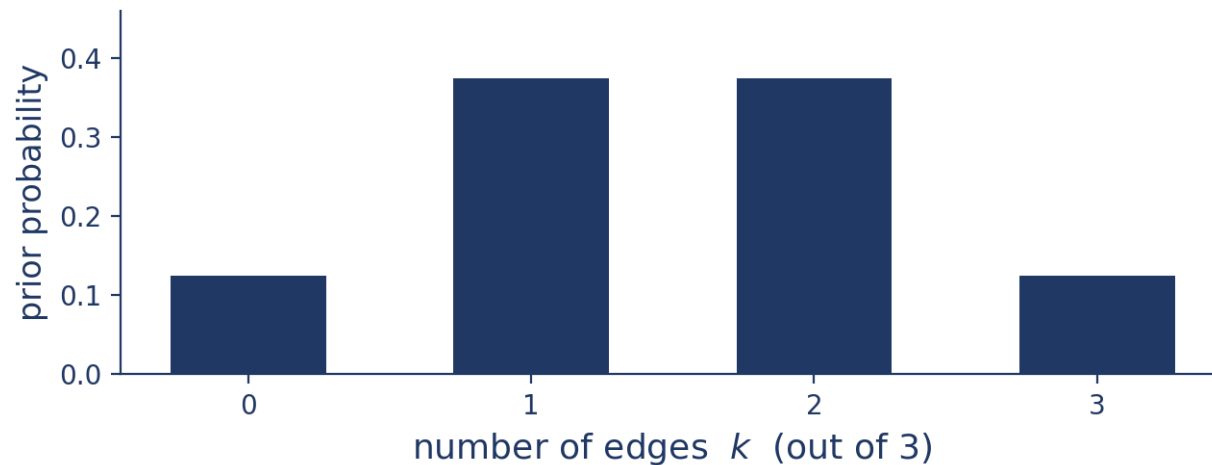
**Beta–Bernoulli** — a prior on the overall network density.

# The Bernoulli prior

Each edge is included independently with a fixed probability  $\pi$  (default  $\pi = 0.5$ ).

With  $\pi = 0.5$  every network structure is equally likely a priori.

But the **density** is then binomial: most prior mass sits near half the edges, so very sparse or very dense networks are unlikely.

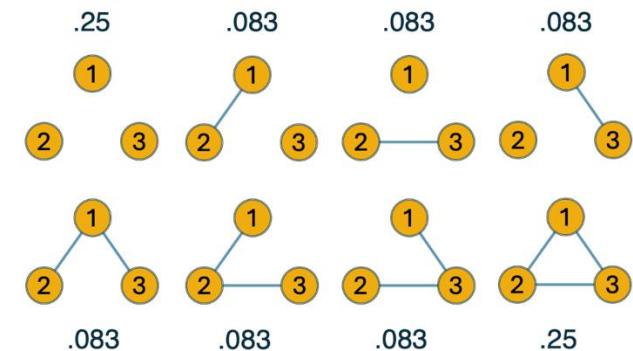
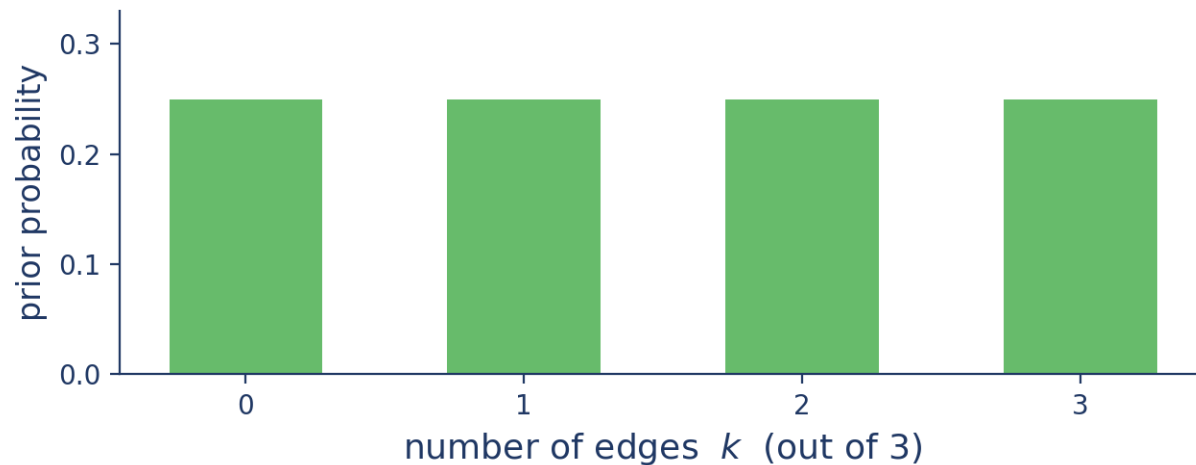


# The Beta–Bernoulli prior

Instead of fixing  $\pi$ , place a **Beta( $\alpha$ ,  $\beta$ )** prior on it

**Beta(1, 1)** makes every density equally likely (uniform over the number of edges).

This automatically **corrects for multiple testing**.





# What they cannot express

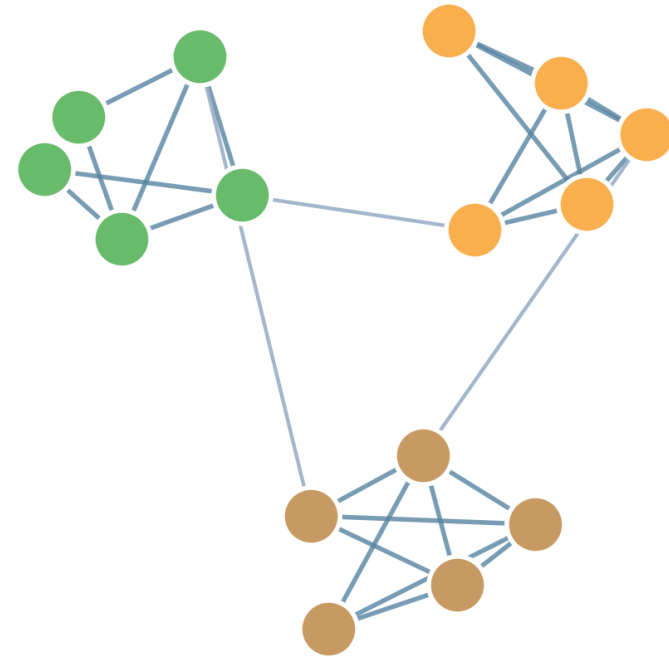
Neither prior can say that the variables form **clusters**.

Yet clustering is exactly what many psychological theories predict.

# Clustering in psychological networks

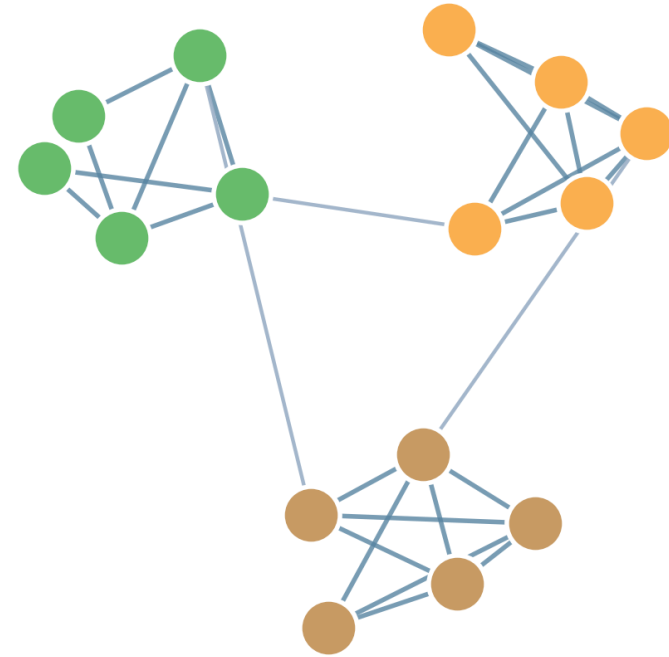
Intelligence, personality, and psychopathology all assume variables group into **clusters** (communities).

Connections are **dense within** a cluster and **sparse between** clusters — sometimes linked by a few bridge nodes.



# What is a cluster?

A **cluster** (block / community) is a set of nodes that are more densely connected to each other than to the rest of the network.



# Two ways to find clusters

**Post-hoc:** first estimate the network, then run a community-detection algorithm on it (e.g. EGA).

**Model-based:** build the clustering assumption into the prior itself and then estimate and test it.

The Stochastic Block Model takes the second route.

# The Stochastic Block Model (SBM) prior

Put the clustering assumption directly into  $\mathbf{p}(\mathbf{G})$ .

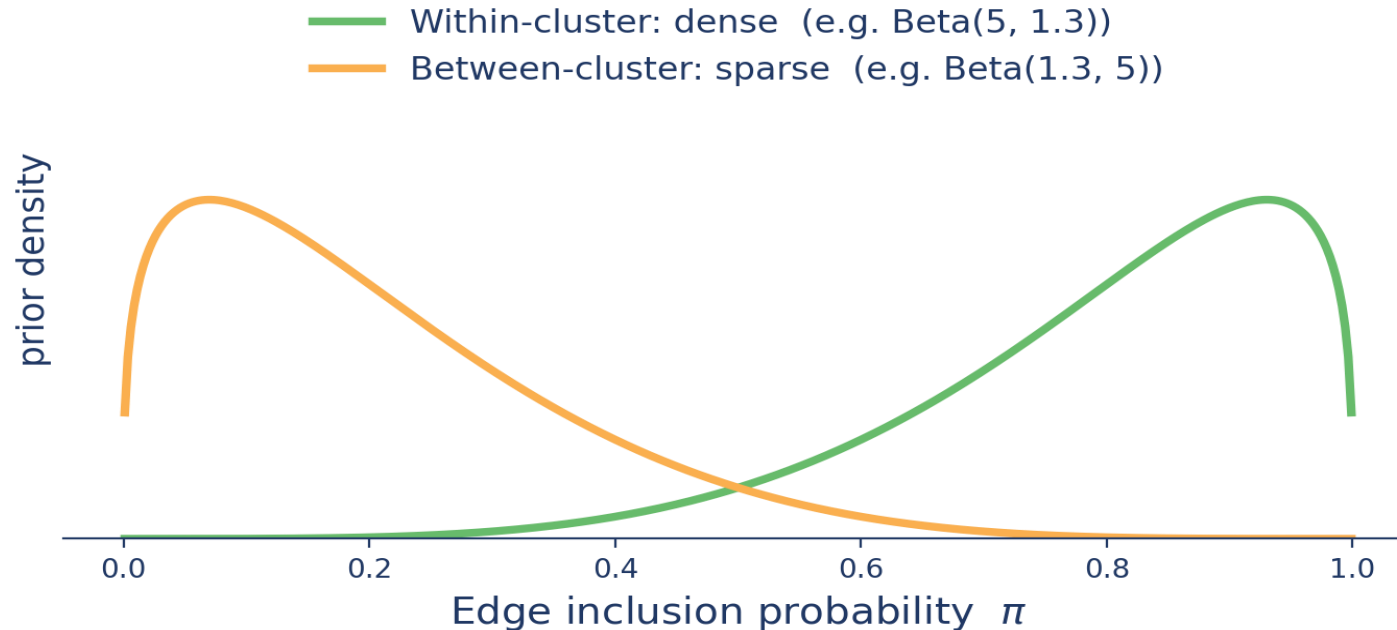
Each node belongs to one of **B latent blocks**; the label of node  $i$  is  $z_i$ .

The probability of an edge depends only on the blocks of the two nodes:

$$P(\text{edge between } i, j) = \pi_{z_i z_j}$$

# Dense within, sparse between

Two separate **Beta–Bernoulli** priors on edge inclusion:  
within a block — edges are more likely;  
between blocks — edges are less likely.



# The SBM **generalizes** the Beta-Bernoulli

With a single block (**B = 1**), every edge is within-cluster, so the SBM prior is exactly the Beta–Bernoulli prior.

With  $B > 1$  it adds the possibility of multiple clusters

$$B = 1 \implies \text{SBM prior} = \text{Beta–Bernoulli prior}$$

# The number of clusters $B$

Usually we do not know the number of clusters in advance.

So the block labels  $\mathbf{z}$  and the number of blocks  $\mathbf{B}$  become parameters, estimated from the data.

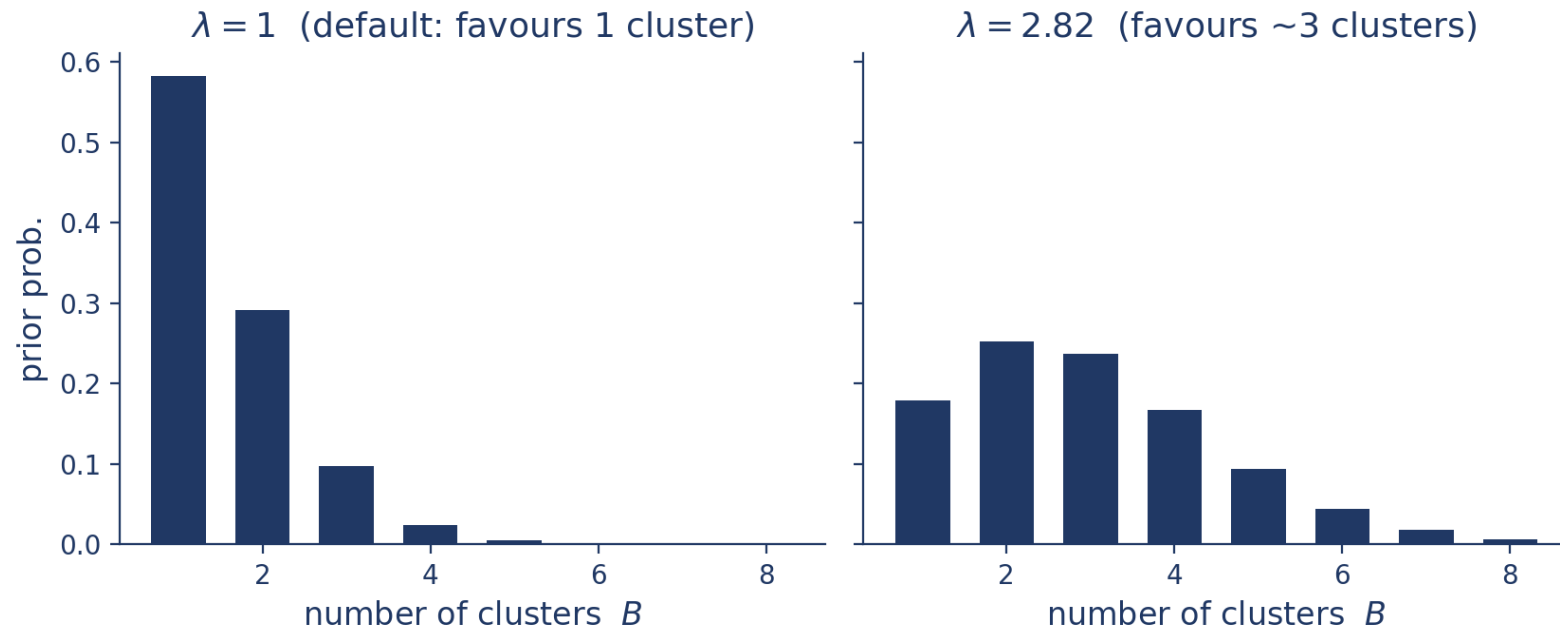
We place a prior on  $B$  (a truncated Poisson):

$$B \sim \text{Poisson}^+(\lambda), \quad B \geq 1$$



# Hyperparameter 1: how many clusters?

$\lambda$  is the rate of the truncated Poisson on  $B$ : it controls the expected number of clusters a priori.



# Hyperparameter 1: how many clusters?

$\lambda$  is not exactly the expected number of clusters, but tracks it closely:

$\lambda = 1 \rightarrow \sim 1$  cluster (default)

$\lambda = 1.59 \rightarrow \sim 2$  clusters

$\lambda = 2.82 \rightarrow \sim 3$  clusters

$\lambda = 3.92 \rightarrow \sim 4$  clusters

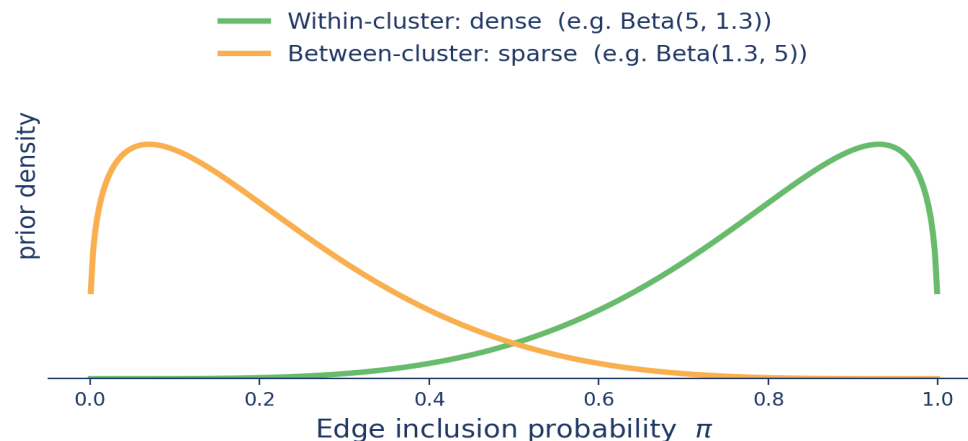
For 5 or more, set  $\lambda$  to the expected number.

# Hyperparameter 2: within / between density

A Beta prior for **within-block** edges and a separate Beta prior for **between-block** edges.

Make within-block inclusion lean toward 1 and between-block lean toward 0 to encode dense-within, sparse-between.

Defaults are uniform, Beta(1, 1), for both.



# Hyperparameter 3: cluster-size balance

A **Dirichlet( $\epsilon$ )** prior controls how evenly nodes spread across clusters.

Large  $\epsilon \rightarrow$  similar-sized clusters; small  $\epsilon \rightarrow$  some large, some tiny.

Default  $\epsilon = 1$  is neutral

# From prior to clustering analysis

With the SBM prior, clustering emerges during estimation. We can now:

- estimate the **number of clusters**,
- test** whether there is clustering at all,
- estimate **which nodes** cluster together.

# Output 1: posterior over the number of clusters

How many clusters does the data support? Here, most posterior mass is on  **$B = 2$** .



## Output 2: testing for clustering

The **clustering Bayes factor** compares two hypotheses:

$$\mathcal{H}_1 : B > 1 \text{ (clustering)} \quad \text{vs.} \quad \mathcal{H}_0 : B = 1 \text{ (no clustering)}$$

$$\text{BF}_{10} = \frac{p(\mathcal{H}_1 | X) / p(\mathcal{H}_0 | X)}{p(\mathcal{H}_1) / p(\mathcal{H}_0)}$$

## Output 2: testing for clustering

**BF** > 1 → evidence for clustering; **BF** < 1 → evidence against.

We can also compare two specific counts, e.g. **B = 2 vs B = 3** (a point hypotheses Bayes factor).

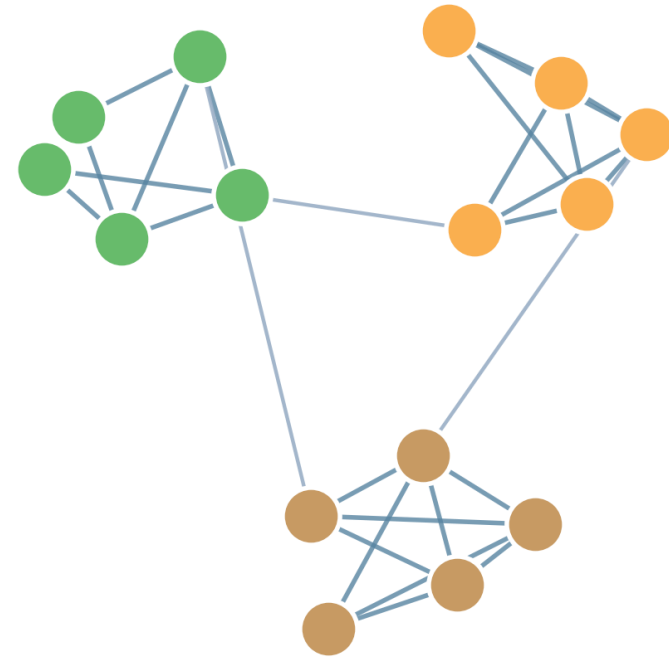
Useful when theory predicts a particular number of clusters.



# Output 3: which nodes cluster together?

Each MCMC iteration assigns nodes to blocks.

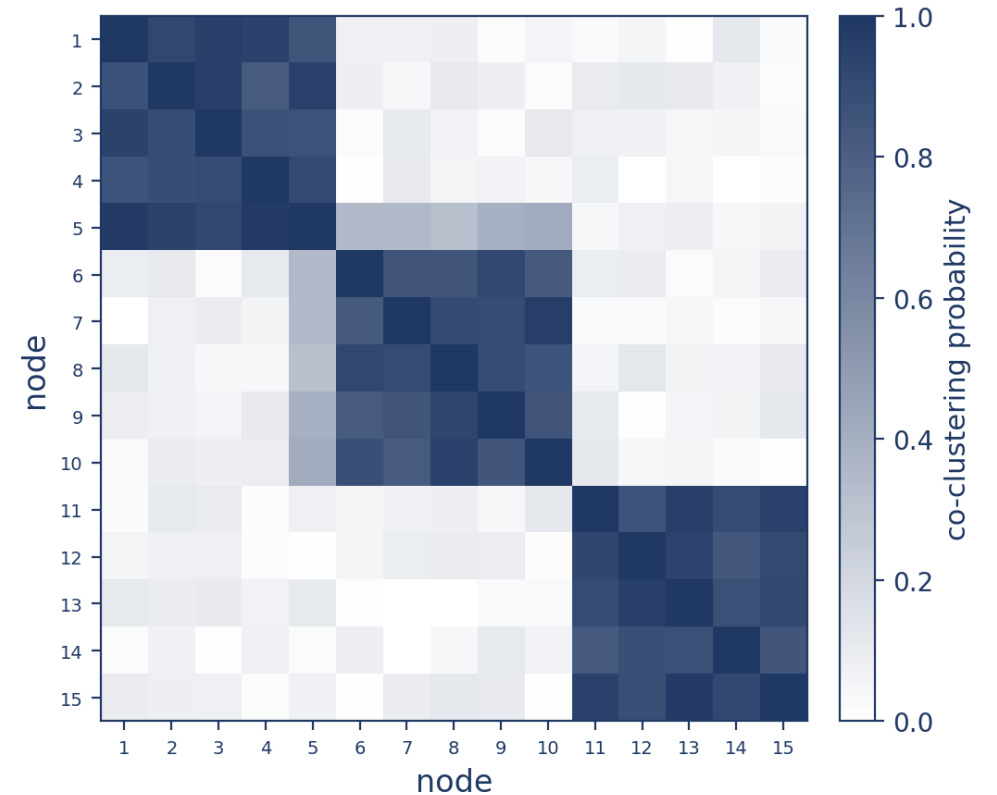
We summarize with the posterior **mode** and posterior **mean** allocation.



# Output 3: the co-clustering matrix

For each pair of nodes: the proportion of iterations they shared a cluster.

Clear blocks = well-supported clusters; diffuse rows = **unstable** nodes (e.g. node 5).



# Take-home: clustering as a testable hypothesis

The SBM prior turns a substantive theory (variables form clusters) into part of the statistical model.

We can **estimate** the clusters and **test** whether they exist, not just detect them post-hoc.

# Bonus: where do the hyperparameters come from?

Choosing  $\lambda$  and the Beta shapes can be hard.

Recent work explores using **large language models** to elicit these priors from a description of the items and theory.

See: Sekulovski, Waaijers & Arena (2025), LLM-Based Prior Elicitation for Bayesian Graphical Modeling.

# Session setup



Part I: Theory



Part II: Practical



Part III: Project